

4DPAPERS

v0.1.2

WRITE · EMBED FIGURES · SHIP ONE FILE

# 4Dpapers

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One file. Any browser. **Still alive.**

CARDIAC EP / CFD / FEA / GEOMETRY

ISOCHROME · 4D FIELD

swipe →

## THE PROBLEM

# Your results die in the PDF.

You run the simulation — a beating heart, a turbulent flow, a loaded part — then flatten all that 3D and time into **one dead image**. Or you hide it behind a platform login.

## WHAT IT IS

# A research-paper IDE.

You write the **whole paper** — prose, equations, references — and drop interactive 3D + time figures straight into the text.

paper.qmd

```
## Ventricular activation
```

```
The depolarization wavefront sweeps  
through the tissue...
```

```
{{< 4d-image src="heart.vtu" field="Vm" >}}
```

## THE DIFFERENCE

# A document, not a dependency.

paper.html

## 3· Ventricular activation

$$\partial V_m / \partial t = \nabla \cdot (D \nabla V_m) - I_{ion}$$



FIGURE 1 · LIVE  
rotate · play · switch field



any browser

offline

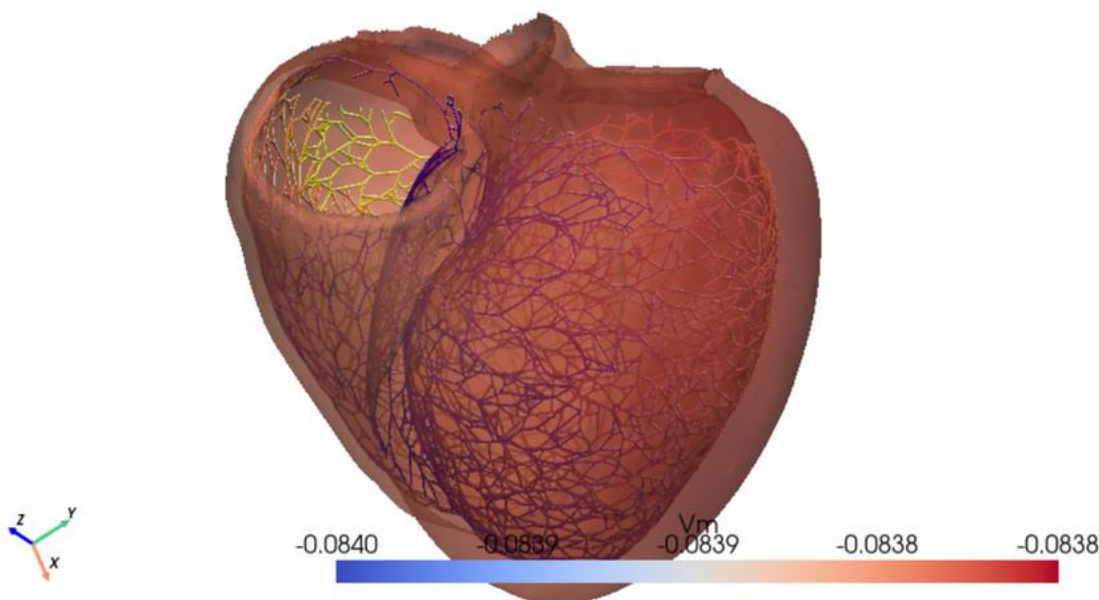
email · USB

Others host your figure on **their server**.  
4Dpapers seals the whole paper into **one file**.

P R O O F

# Real science. Still interactive.

▶ heart.vtu ·  $V_m$  ·  $t = 0.12$  s



Cardiac electrophysiology — rotate the heart, switch fields, scrub the activation in time. In the browser. Offline.

NO LOCK-IN

# Reads what you already have.

OpenFOAM

VTK

VTU

PVD

Exodus

XDMF

CGNS

EnSight

STL

OBJ

PLY

Gmsh

+ 27 formats

Compiles to interactive **HTML** and a clean **PDF** — and the whole thing runs **fully offline**.

GET STARTED

# Run it in one command.

```
IMAGE=ghcr.io/4dpapers/4dpapers:0.1.2  
FOURD_WORKSPACE=./examples/niederer docker compose up
```

→ [github.com/4dpapers/4dpapers](https://github.com/4dpapers/4dpapers)

SOURCE-AVAILABLE · RUNS OFFLINE · v0.1.2